LEC6 - Additional Properties of Probability

Theorem 15: Continuity Theorem of Probability

Let $\{C_n\}$ be an increasing sequence of events. Then

$$\lim_{n\to\infty} P(C_n) = P\left(\lim_{n\to\infty} C_n\right) = P\left(\bigcup_{n=1}^{\infty} C_n\right)$$

Let $\{C_n\}$ be a decreasing sequence of events. Then

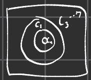
$$\lim_{n\to\infty} P(C_n) = P\left(\lim_{n\to\infty} C_n\right) = P\left(\bigcap_{n=1}^{\infty} C_n\right)$$

Proof for increasing

Suppose $\{C_n\}$ is an increasing sequence of events

$$\bigcup_{n=1}^{\infty} C_n = \lim_{n\to\infty} C_n \Rightarrow P\left(\bigcup_{n=1}^{\infty} C_n\right) = P\left(\lim_{n\to\infty} C_n\right)$$

We want to show $\lim_{n\to\infty} P(C_n) = P\left(\bigcup_{n=1}^{\infty} C_n\right)$



$R_1 = C_1$

$R_2 = C_2 - C_1 = C_2 \cap C_1^c$

$R_3 = C_3 - C_2 = C_3 \cap C_2^c$

$\vdots$

$R_n = C_n - C_{n-1} = C_n \cap C_{n-1}^c$

Here R_n are pairwise mutually exclusive

$$\bigcup_{n=1}^{\infty} R_n = \bigcup_{n=1}^{\infty} C_n$$

$$\Rightarrow P\left(\bigcup_{n=1}^{\infty} C_n\right) = P\left(\bigcup_{n=1}^{\infty} R_n\right) \overset{A3}{=} \sum_{n=0}^{\infty} P(R_n)$$

telescoping sum

$$\Rightarrow \lim \left[P(C_1) + \left[P(C_2) - P(C_1) + P(C_3) - P(C_2) \cdots\right]\right]$$

$$= \lim_{n\to\infty} P(C_n)$$

More Properties

- $P\left(\bigcap_{n=1}^{\infty} C_n\right) = P\left(\left(\bigcup_{n=1}^{\infty} C_n^c\right)^c\right) = 1 - P\left(\bigcup_{n=1}^{\infty} C_n^c\right)$

- $P(A-B) = P(A) - P(A\cap B)$

- $\underbrace{\min(P(A), P(B))}_{\substack{A \subset A\cup B \\ B \subset A\cup B}} \le P(A\cup B) \le P(A) + P(B)$

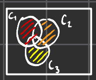
Theorem 16: Boole's Inequality

Let $\{C_n\}$ be an arbitrary sequence of events. Then

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) \le \sum_{n=1}^{\infty} P(C_n)$$

- Upper bound for union

Proof:



$R_1 = C_1$

$R_2 = C_2 - C_1$

$R_3 = C_3 - (C_1 \cup C_2)$

Mutually exclusive & $\bigcup_{n=1}^{\infty} R_n = \bigcup_{n=1}^{\infty} C_n$

$R_n \subset C_n \Rightarrow P(R_n) \le P(C_n)$

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) = P\left(\bigcup_{n=1}^{\infty} R_n\right) \overset{\text{axiom 3}}{=} \sum_{n=1}^{\infty} P(R_n) \le \sum_{n=1}^{\infty} P(C_n)$$, as required